String Sizing Justification – 18 / 19 Modules per String

Kampala Marriott Hotel · 185.4 kWp · 299 × Astronergy ASTRO N7 620 W · 2 × Sigenergy Sigen PV 80M1 · Ref. NSP-261105-KMH

1. Design question

For 1,100 V C&I string inverters, 16–17 modules per string is a common rule of thumb. This design uses **19 modules (INV-01: 8 × 19; INV-02: 3 × 19)** and **18 modules (INV-02: 5 × 18)** – 16 strings in total. This note checks that 18–19 modules stay inside every inverter limit and explains why they are preferred here.

2. Data used

Module (datasheet)	Value	Inverter Sigen PV 80M1 (datasheet)	Value
Voc / Vmp (STC)	49.04 V / 41.56 V	Max. DC input voltage	1,100 V
Isc / Imp (STC)	16.11 A / 14.92 A	MPPT voltage range	160 – 1,000 V
Temp. coeff. Voc / Isc	−0.25 %/°C / +0.043 %/°C	Nominal DC voltage / start-up	600 V / 180 V
Temp. coeff. Vmp (assumed)	−0.27 %/°C	MPP trackers × strings per MPPT	6 × 2
Max. system voltage	1,500 V DC	Max. input / short-circuit current per MPPT	40 A / 60 A

Site temperatures (Kampala, 1,180 m): minimum cell temperature 10 °C (cool clear dawn – record ambient minimum ≈ 10–12 °C); maximum cell temperature 70 °C (hot roof, low tilt). Bifacial rear-side current gain taken as +10 %.

3. Voltage check per string length

Modules in series	Voc at 10 °C (≤ 1,100 V)	Margin to 1,100 V	Vmp at 10 °C (≤ 1,000 V)	Vmp at STC	Vmp at 70 °C (≥ 180 V)	Vmp STC ÷ 600 V nominal
16	814 ✔	26 %	692 ✔	665	584 ✔	1.11
17	865 ✔	21 %	735 ✔	707	621 ✔	1.18
18	**916 ✔**	**17 %**	**778 ✔**	**748**	**657 ✔**	**1.25**
19	**967 ✔**	**12 %**	**822 ✔**	**790**	**694 ✔**	**1.32**

Limits for this module and inverter at Kampala temperatures: **maximum 21 modules** (Voc cold ≤ 1,100 V) and **minimum 5 modules** (hot Vmp ≥ start-up voltage). **18 and 19 modules are well inside this range**: the coldest open-circuit voltage of a 19-module string is 967 V – 12 % below the 1,100 V limit – and the operating voltage stays inside the 160–1,000 V MPPT window at all temperatures (694–822 V for 19 modules, 657–778 V for 18 modules).

4. Current check per MPPT (2 strings in parallel)

Check	Value	Limit	Result
Operating current, 2 × Imp	29.8 A (32.8 A with +10 % bifacial gain)	40 A max. input	✔
Short-circuit current, 2 × Isc at 70 °C × 1.10 bifacial	36.1 A	60 A max. short-circuit	✔
Strings per inverter	8 (uses 4 of 6 MPPTs, 2 strings each)	12 strings	✔

String length does not change the current – the current check is the same for 16–19 modules.

5. Comparison of stringing options for 299 modules

Item	Design: 19 / 18 modules	Alternative: 17 / 16 modules
Strings	**16** (11 × 19 + 5 × 18)	18 (11 × 17 + 7 × 16)
Per inverter	8 + 8 strings → 4 MPPTs fully paired on each	9 + 9 strings → 5 MPPTs, one MPPT with a single string
DC isolators + DC SPDs	16 + 16	18 + 18 (+ 4 units)
String home-run cable pairs	16	18 (≈ 12 % more DC cable)
String operating voltage (STC)	748–790 V	665–707 V
DC ohmic loss (same power, same cable)	lower – $\propto 1/V^2$; ≈ 20 % less than 17/16	higher
Inverter efficiency (nearer 600 V nominal)	very slightly lower (typ. < 0.2 %)	very slightly higher
Voc margin to 1,100 V at 10 °C	12 % (19 mod.)	21 % (17 mod.)

6. Why 16–17 is the usual rule – and why it does not limit this site

- 16–17 modules keep the cold open-circuit voltage far below 1,000–1,100 V in **cold climates** (design minimum −10 °C or lower), and keep the operating voltage close to the inverter's 600 V nominal point.
- Kampala is equatorial at 1,180 m – cell temperature does not fall much below 10 °C – so even 19 modules leave a 12 % voltage margin.
- 299 modules divide neatly into **16 strings of 19/18**, pairing exactly on 4 MPPTs per inverter. 16/17-module strings need 18 strings and leave one MPPT on each inverter with a single string.
- Fewer, higher-voltage strings mean fewer isolators and SPDs, less DC cable and lower DC losses, which outweighs the small efficiency advantage of operating closer to 600 V.

> **Conclusion:** 18- and 19-module strings are within the Sigen PV 80M1 limits at all Kampala temperatures (maximum 21 modules) and are the better fit for 299 modules. The design is kept at 16 strings: INV-01 8 × 19; INV-02 3 × 19 + 5 × 18. A 17/16 layout is also technically possible if a client specification requires it.

Action: the PVsyst model currently uses 23 strings × 13 modules – update it to the design stringing above before the final simulation. Vmp temperature coefficient assumed (−0.27 %/°C) – confirm with the module manufacturer's datasheet or PAN file.